Brady Bangasser

HPC & Systems Engineer

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Systems engineer running production HPC clusters and researching high-performance and distributed computing, from scheduler internals to GPU-offload optimization at supercomputer scale.

EXPERIENCE

Research Assistant — Compiler Optimization · Iowa State University — Aug 2025 – Present

Ames, IA

- Built OMPRTF (OpenMP RunTime Fabric), a dynamic-analysis pipeline that applies aggressive, conditional optimizations to OpenMP GPU-offload programs by lowering them to LLVM IR.
- Wrote an LLVM optimization pass that breaks opaque OpenMP runtime calls into discrete IR instructions, pruning redundant host-GPU data movement without destabilizing kernels.
- Built a profiling library that correlates OpenMP target logs with IR metadata to surface duplicate transfers and unused memory automatically.
- Co-developed an LLVM-based tool that restructures disk I/O access patterns to speed up data loading for large ML/AI workloads.
- Analyzed Darshan I/O logs from Iowa State's Nova cluster and the Polaris supercomputer to locate and resolve system-level data-loading bottlenecks.

System Administrator · Iowa State University — Apr 2025 – Present

Ames, IA

- Operate and maintain a production HPC cluster with SLURM, Kubernetes, Active Directory, Ansible, Prometheus, and Apptainer, keeping it secure and available for a large user base.
- Built diagnostic and monitoring workflows that isolate hardware degradation, network bottlenecks, and software-stack faults before they take nodes offline.
- Automated node provisioning and recovery with Ansible so failed or new nodes rejoin service quickly.
- Validated cluster performance and hardware integrity with distributed LINPACK and NCCL benchmarks, stress-testing multi-GPU InfiniBand bandwidth.
- Developed and tested a new feature in a customized SLURM scheduler in C, with Makefile-driven builds, debugging, and validation for stability and compatibility.

Research Assistant — Machine Learning · Iowa State University — Oct 2024 – May 2025

Ames, IA

- Designed and trained object-classification models with PyTorch and OpenCV, tuning data preprocessing and tensor operations to cut training latency without losing accuracy.
- Benchmarked accuracy and throughput across GPU configurations to guide training choices.
- Tested perception models on self-driving vehicles to catch false and failed classifications.

Research Assistant — High Performance Computing · Bethel University — Aug 2023 – Aug 2024

Saint Paul, MN

- Built and configured two bare-metal HPC clusters from scratch: networking, custom DNS and routing, user management, and SLURM job scheduling with VMware and Debian.
- Implemented a wildfire simulation in OpenMP, CUDA, C, and Fortran to benchmark cluster performance across heterogeneous compute.
- Wrote a lightweight daemon and control program to manage per-node status lighting across the clusters.

EDUCATION

M.S. Computer Science · Iowa State University — Aug 2025 – May 2027

GPA: 4.00

- Concurrent B.S./M.S. program
- Coursework: Advanced Topics in High Performance Computing, Distributed Systems and Middleware, Data Security in Machine Learning, Network Design and Security

B.S. Computer Science · Iowa State University — Aug 2024 – Dec 2026

GPA: 3.79 · Dean's List · Minors in Cybersecurity and Data Science

- Coursework: Compiler Design and Development, AI and Machine Learning in High Performance Computing, Network Architecture and Security, Theory of Computing, Software Design Practices

TECHNICAL SKILLS

Languages: C, C++, Python, Go, Rust, Bash, CUDA, Fortran
Infrastructure & Platform: Kubernetes, SLURM, Apptainer, Linux/Unix, Git
Compilers & HPC: LLVM, OpenMP, OpenMPI, PyTorch